Statement of Work
Fort Belvoir
Single Award Task Order Contract(SATOC)
Indefinite Delivery/Indefinite Quantity (ID/IQ)

As of 19 September 2024

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1.0 CONTRACT DESCRIPTION

This is a Firm Fixed Price (FFP) Indefinite Delivery-Indefinite Quantity (IDIQ) Single Award Task Order (SATOC) contract for roofing services as defined under NAICS 238160. The Contractor shall provide all personnel, equipment, supplies, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform the SATOC as defined in this Statement of Work (SOW). The Contractor shall perform to the standards in this contract.

1.1 GENERAL REQUIREMENTS:

This is a non-personal services contract to provide Single Award Task Order (SATOC) Indefinite Delivery/Indefinite Quantity (ID/IQ) procurement options to Fort Belvoir (Installation) and satellite camps/facilities. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.

1.2 DESCRIPTION OF SERVICES/INTRODUCTION:

The Contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform the JOC as defined in this Performance Work Statement (PWS), except for those items specified as government furnished property and services. The Contractor shall perform to the standards in this contract.

2.0 OBJECTIVE

This contract is for a broad range of maintenance, repair, sustainment, and minor construction projects on real property at Fort Belvoir. During the contract period, the Government will identify roofing tasks required to complete each specific project. These tasks will include roof removal, installation, repair, sealing and or replacements to buildings.

The Contractor shall furnish site work as required per the line item in the Construction Task Catalog (CTC), commonly referred to as a Unit Price Book (UPB). The Contractor shall use the site-specific Gordian Construction Task Catalog as the basis for pricing all task order for this contract. The site-specific Gordian catalog is specifically formulated for the Ft. Belvoir area. The Contractor shall perform in accordance with (IAW) local policies, National Fire Protection Association (NFPA) codes, and industry and trade organization standards.

The services covered by this contract are for real property repair and construction. Construction and repair include, but are not limited to, the following systems and disciplines:

Asphalt roof shingle installation	Painting, spraying, or coating, roof	Roofing, built-up tar and gravel, installation	Steep slope roofing installation
Copper roofing installation	Roll roofing installation	Shake and shingle, roof, installation	Treating roofs (by spraying, painting or coating)

Corrugated metal roofing installation	Roof membrane installation	Sheet metal roofing installation	
Galvanized iron roofing installation	Roof painting, spraying, or coating	Skylight installation	
Low slope roofing installation	Roofing contractors	Solar reflecting coating, roof, application	

The Government will implement the processes set forth in the Scope of Work (SOW) to develop and issue individual Task Orders. Individual SOWs will vary in size depending on the nature of the work. The Contractor shall provide sufficient technical support, project management, accurate estimates, and accommodate a number of concurrent active projects. The Contractor shall handle rapid increases in work volume, perform according to schedule without delay, and complete submittals and payrolls in a timely manner.

3.0 SCOPE: The Contractor shall provide all supervision, plant, labor, equipment, materials and appliances necessary to perform all operations in connection with the repair and replacement of existing roofs on various building structures located at Fort Belvoir, in accordance with this Statement of Work, specifications and drawings, and subject to the terms and conditions of the contract. Completed work shall result in each building or facility being watertight and impervious to the weather and its elements.

4.0 GENERAL: All work shall be completed by personnel skilled in the type of work involved. When new work adjoins, connects or abuts existing work or facilities, the existing work or facilities shall be altered as required and the connections made in an approved, workmanlike and professional manner. All existing work or facilities damaged by the Contractor's operations, other than facilities specified to be removed in a task order, shall be repaired or replaced by the Contractor at no cost to the Government. Prior to performance of work, a preconstruction orientation shall be held at Fort Belvoir unless waived by the Government. The Contractor shall have a representative attend who has authority to make decisions for and sign contractual documents on behalf of the Contractor. The Contractor must have or provide the following:

a. Hold and maintain a Virginia Class A Contractor's License.
b. Have a local office within two (2) hours of Ft. Belvoir for on-site management.
c. Provide contact information for vetted subcontractors, if applicable.

5.0 LOCATION: The site of the proposed work is located in Fort Belvoir, Virginia and Area Garrisons which include Fort Walker (fka Fort A.P. Hill), Joint Base Myer Henderson Hall (JBMHH), Rivanna Station, Arlington National Cemetery, and Fort McNair.

6.0 PRINCIPAL FEATURES: Each task order issued by the Government under this contract shall specifically identify the building(s) to receive roof repair and/or replacement work along with the associated Pricing Schedule items and quantities necessary to complete each building roof or portion thereof. Although the contract does not explicitly identify individual buildings, roofs or other structures to be constructed, repaired or replaced, the contract bid schedule is structured to order quantities of materials, rather than buildings, sufficient to complete all work required to provide a completed project. It is the Government's intent that the Pricing Schedule items and quantities identified in a task order, with respect to a particular building, shall be sufficient to complete all work related to that building. All Pricing Schedule items and quantities of material ordered to complete a particular building, to include within scope task order modifications, are a bona fide requirement of the Fiscal Year in which the task order is placed.

Unified Facilities Guide Specifications (UFGS) referenced within this Statement of Work is available at: https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs

Unified Facility Criteria (UFC) documents referenced within this Statement of Work is available at:

https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc

Roof work shall be performed on "BUILDINGS", which include warehouses, administrative and operation buildings, shops, barracks, mess halls, and other miscellaneous structures on Fort Belvoir. The work performed will include Pricing Schedule items in accordance with the contract documents, specifications, and drawings.

7.0 RATE OF WORK PERFORMANCE:

The period of time, in calendar days, allowed for completion of each task order shall be as follows: counting of calendar days allowed for each task order shall start from the *date of notice to proceed* of the task order. The time allowed for submittals shall be added to the time allowed for the first task order.

Each Task Order	Calendar Days to Complete
Under $10,000	20
$10,000 to $19,999	30
$20,000 to $99,000	45
Over $100,000	Will be negotiated

Note: Contractor's rate of performance for multiple task orders issued at the same time (within one week) shall be summed for aggregate total. This aggregate total will be compared to the rate of performance table in the contract for the number of days allowed to complete a group of task orders issued at the same time. For example, four task orders issued at same time worth $10,000 each for a total of $40,000 are given 45 days for completion of all four task orders.

8.0 SUBMITTALS:

Submittals shall be provided as required by the contract. Failure to furnish submittals may result in a suspension of work. Materials furnished and/or installed that do not meet the specifications may be rejected by the Contracting Officer or his/her duly appointed representative. The following electronic submittals shall be provided after contract award for review and approval.

Submittal Date Due

Anti-Terrorism (AT) Level 1 Training	60 calendar days after contract award
TARP Training	60 calendar days after contract award
OPSEC Training	60 calendar days after contract award
Insurance Certificate	30 calendar days after contract award
Generic Site Safety & Health Plan	30 calendar days after contract award
Environmental Protection Plan	30 calendar days after contract award

Contractor Quality Control Plan	30 calendar days after contract award
Waste Management Plan	30 calendar days after contract award
Material (all materials required for the contract)	30 calendar days after contract award

9.0 SECURITY REQUIREMENTS

Contract Clause FAR 52.204-2 Security Requirements and Alternate II and the following apply:

a. The Contractor shall ensure that all contractor and sub-contractors employees who require physical access to Army Bases provide all information required for a check of records through databases including, but not limited to, the National Crime Information Center (NCIC) Interstate Identification Index (III) and the Terrorist Screening Database. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at the installation change, the Government may require changes in contractor security matters or processes.

b. All contractor and associated sub-contractor employees who require physical access to Army Bases for contract performance shall present an Acceptable Identity Source Document, as identified in Section 2 of Attachment 4 of DoD Directive-Type Memorandum 09-012, "Interim Policy Guidance for DoD Physical Access Control" (available at http://www.dtic.mil/whs/directives/corres/pdf), filename "DTM-09-012.pdf" to access the installation.

c. The Contractor shall maintain awareness of all Prime and Sub-Contractor employees who are vetted and approved for access onto the Base. Contractor shall notify the KO and COR in writing within 48 hours of any contractor or sub-contractor employees who resigns, is terminated, or for any other reason and no longer authorized to access Army Base for the purpose of performance of this contract. For contracts with a performance period longer than three months, the Contractor shall reconcile his roster of approved personnel with the Base Visitor Control Center (VCC) to ensure that both the Contractor and Contract Officer Representative maintain accurate awareness of the contractor and subcontractor employees who are authorized to access the installation.

   i. All contractor and all associated sub-contractors employees requiring physical access to Army Base for contract performance shall be pre-screened by the Contractor using the E-verify Program at http://www.dhs.gov/E-Verify website to meet the established employment eligibility requirements and shall provide only eligible employees for contract performance while on the installation.

   ii. All contractor and all associated sub-contractors employees requiring physical access to Army Base for contract performance shall complete AT Level I awareness training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the COR or to the contracting officer if a COR is not assigned, within 10 calendar days after completion of training by all employees and subcontractor personnel.

AT level I awareness training is available at https://jkodirect.jten.mil/Atlas2/faces/page/login/Login.seam .

iii. All contractor and sub-contractor employees shall attend a briefing on the local iWATCH program provided by the JBM-HH ATO or Security Office. This locally-developed briefing will be used to inform employees of the types of suspicious behavior to report to the COR or the Base Police Department. This briefing shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees' commencing performance with the results reported to the COR no later than 30 days after contract award and new employees' commencing performance.

iv. All contractor and all associated sub-contractors employees requiring physical access to Army Base for contract performance shall complete Operations Security (OPSEC) training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the COR or to the contracting officer if a COR is not assigned, within 10 calendar days after completion of training by all employees and subcontractor personnel. OPSEC training is available at http://cdsetrain.dtic.mil/opsec/ .

v. The Contractor shall ensure that all contractor and sub-contractors employee Common Access Cards (CACs), badges, passes or other access credentials which were issued under this contract are surrendered to the COR upon completion of the contract. CACs, badges, passes and other access credentials are DoD property and failure to surrender them may be punishable under 18 USC § 701.

10. SUPERVISION

10.1 Minimum Communication Requirements

The Contractor shall have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the jobsite at all times during the performance of contract work. In addition, if a Quality Control (QC) representative is required on the contract, then that individual must also have fluent English communication skills.

10.2 Superintendent Qualifications

The project superintendent must have a minimum of 6-years of experience in construction with at least 3 of those years as a superintendent on projects similar in size and complexity of the task order. The individual must be familiar with the requirements of EM 385-1-1 (current version) and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project.

10.3 Duties

The project superintendent is primarily responsible for managing and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of the contract until the work is completed and accepted.

10.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

10.5 Preconstruction Conference

After award of the contract but prior to commencement of any work at the site, a meeting with the Contracting Officer will be conducted to discuss and develop a mutual understanding relative to the administration of the value engineering and safety program, preparation of the schedule of prices or earned value report, shop drawings, and other submittals, scheduling programming, prosecution of the work, and clear expectations of the "Interim DD Form 1354" Submittal. Major subcontractors who will engage in the work must also attend.

10.6 Electronic Mail (Email) Address

The Contractor must establish and maintain electronic mail (e-mail) capability along with the capability to open various electronic attachments as text files, pdf files, and other similar formats. Within 10 days working days after contract award, the contractor shall provide the Contracting Officer with a single (one only) e-mail address for electronic communications from the Contracting Officer related to this contract including, but not limited to contract documents, invoice information, request for proposals, and other correspondence. The Contracting Officer may also use email to notify the Contractor of base access conditions when emergency conditions warrant, such as hurricanes or terrorist threats. **Multiple email addresses are not allowed.**

It is the Contractor's responsibility to make timely distribution of all Contracting Officer initiated e-mail with its own organization including field office(s). Promptly notify the Contracting Officer, in writing, of any changes to this email address.

11.0 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN.

11.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

11.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

11.3 Correction of Deficiencies

In the event that a deficiency is found, additional measures shall be taken, as deemed necessary by the Contracting Officer, to determine the extent of the deficiency. Corrective actions shall be taken as directed by the Contracting Officer to correct the deficiency.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10% (ten percent) of the pay request amount from each payment period until such revisions to the project schedule have been made.

12.0 PROJECT SCHEDULE DETAILED REQUIREMENTS

12.1 Level of Detail Required

The Contractor shall develop the Project Schedule to show the appropriate level of detail to address all major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an adequate level of detail will result in its disapproval by the Contracting Officer and necessitate a meeting with the Contractor, requirement owner, and Contracting.

12.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Less than 2% (two percent) of all non-procurement activities may have Original Durations (OD) greater than 20 work days or 30 calendar days.

12.2.a. Mandatory Tasks

Include the following activities/tasks in the initial project schedule and all updates.

i. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).

ii. Submission and approval of as-built drawings.

iii. Submission and approval of DD1354 data and installed equipment lists.

iv. Contractor's pre-final inspection.

v. Correction of punch list from Contractor's pre-final inspection.

vi. Government's pre-final inspection.

vii. Correction of punch list from Government's pre-final inspection.

viii. Final inspection.

12.3 CORRECTION OF DEFICIENCIES

Where any form of deficiency is found, the Contractor shall take additional measures, as deemed necessary by the Contracting Officer, to determine the extent of the deficiency and perform corrective actions as directed by the Contracting Officer.

12.4 Government Activities

The Contractor shall indicate any Government and other agency activities that could impact progress. These activities include, but are not limited to: environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

12.5 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

12.6 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations. Lags must not be in place to artificially absorb float, or to replace proper schedule logic.

a. Leads (negative lags) are prohibited.

b. Start to Finish (SF) relationships are prohibited.

12.7 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

12.8 Percent Complete

The Contractor shall update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete, except for any remaining minor punch list work, and which do not restrain the initiation of successor activities, may be declared 100 percent complete to allow for proper schedule management.

12.9 Remaining Duration

The Contractor shall update the remaining duration for each activity based on the number of estimated work days it will take to complete the activity. Remaining duration may not mathematically correlate with percentage found under paragraph entitled Percent Complete.

12.10 Cost Loading of Closeout Activities

The Contractor shall cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

12.11 As-Built Drawings

If there isn’t a separate contract line item (CLIN) for as-built drawings, the Contractor shall cost load the "Submission and approval of as-built drawings" activity not less than $35,000 or 1 percent of the present contract value, whichever is greater, up to $200,000. The activity will be declared 100 percent complete upon the Government's approval.

12.12 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and manhours) and the Government agreeing that the schedule is reasonable and achievable.

b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

13.0 PROJECT SCHEDULE SUBMISSIONS

13.1 General Information

The Contractor shall provide the submissions as described below. The data CD/DVD, reports, and network diagrams required for each submission are contained in paragraph 13.5 SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, then the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

The Contractor’s review of comments made by the Government on the schedule(s) does not relieve the Contractor from compliance with requirements of the Contract Documents.

13.2 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, the Contractor shall submit the Preliminary Project Schedule, defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. The Contractor shall completely cost load the Preliminary Project Schedule to balance the contract award CLINs shown on the Price Schedule. The Preliminary Project Schedule may be a summary in nature for the remaining performance period. The Preliminary Project Schedule must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. The Government’s acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. The Contractor shall activity code any activities that are summary in nature after the first 90 calendar days with Bid Item (CLIN) code (BIDI), Responsibility Code (RESP) and Feature of Work code (FOW).

13.3 Initial Project Schedule Submission

The Contractor shall submit the Initial Project Schedule for approval within 42 calendar days after the notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represents all work through the entire contract performance period. No payment will be made for work items not fully detailed in the Project Schedule.

13.4 Periodic Schedule Updates

The Contractor shall update the Project Schedule on a regular basis, monthly at a minimum. The Contractor shall provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph 13.6.a PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess the Contractor's progress.

a. The Contractor shall update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. These updates are subject to the approval of the Government at the meeting.

b. AS and AF dates must match the date(s) reported on the Contractor's Quality Control Report for an activity start or finish.

13.5 SUBMISSION REQUIREMENTS

The Contractor shall provide Periodic Schedule Updates throughout the life of the project stating either On-target or Delay. If there is a Delay, then the Contractor shall provide a detailed explanation of the cause for the delay, and how long (in days) the Contractor estimates the delay will postpone the project.

13.6 PERIODIC SCHEDULE UPDATE

13.6.a Periodic Schedule Update Meetings

i. The Contractor shall conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment.
ii. The Contractor shall conduct meetings at least monthly within five days of the proposed schedule data date.
iii. The Contractor shall provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting.
iv. The Contractor's authorized scheduler must organize, group, sort, filter, and perform schedule revisions as needed and review functions as requested by the Contractor and/or Government.
v. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis.
vi. The meeting will last no longer than 8 hours.
vii. The Contractor shall provide a draft of the proposed narrative report and schedule data file to the Government a minimum of two workdays in advance of the meeting.
viii. The Contractor's Project Manager and scheduler must attend the meeting with the authorized representative of the Contracting Officer.
ix. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work.
x. Following the periodic schedule update meeting, the Contractor shall make corrections to the draft submission. The Contractor shall include only those changes approved by the Government in the submission and invoice for payment.

14.0 GOVERNMENTAL SAFETY REQUIREMENTS

14. 1 PART 1 GENERAL: REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred within the text by the basic designation only.

14.2 AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ASSP A10.34	(2001; R 2012) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2014) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2012) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2016) The Fall Protection Code
ASSP Z359.2	(2017) Minimum Requirements for a Comprehensive Managed Fall Protection Program
ASSP Z359.3	(2017) Safety Requirements for Lanyards and Positioning Lanyards
ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ASSP Z359.7	(2011) Qualification and Verification Testing of Fall Protection Products
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses

ASSP Z359.12	(2009) Connecting Components for Personal Fall Arrest Systems
ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems

14.3 ASME INTERNATIONAL (ASME)

ASME B30.3	(2016) Tower Cranes
ASME B30.5	(2014) Mobile and Locomotive Cranes
ASME B30.7	(2011) Winches
ASME B30.8	(2015) Floating Cranes and Floating Derricks
ASME B30.9	(2014; INT Feb 2011 - Nov 2013) Slings
ASME B30.20	(2013; INT Oct 2010 - May 2012) Below-the-Hook Lifting Devices
ASME B30.22	(2016) Articulating Boom Cranes
ASME B30.23	(2011) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; INT Jun 2010 - Jun 2014) Rigging Hardware

14.4 ASTM INTERNATIONAL (ASTM)

ASTM F855	(2015) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

14.5 INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2003) Guide for Protective Grounding of Power Lines
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

14.6 NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2018; TIA 18-1) Standard for Portable Fire Extinguishers

NFPA 51B	(2014) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2017; ERTA 1-2 2017; TIA 17-1; TIA 17-2; TIA 17-3; TIA 17-4; TIA 17-5; TIA 17-6; TIA 17-7; TIA 17-8; TIA 17-9; TIA 17-10; TIA 17-11; TIA 17-12; TIA 17-13; TIA 17-14; TIA 17-15; TIA 17-16; TIA 17-17) National Electrical Code
NFPA 70E	(2018; TIA 18-1; TIA 81-2) Standard for Electrical Safety in the Workplace
NFPA 241	(2013; Errata 2015) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2019) Standard for the Control of Gas Hazards on Vessels

14.7 TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2005G; Add 1 2007; Add 2 2009; Add 3 2014; Add 4 2014; R 2014; R 2016) Structural Standards for Steel Antenna Towers and Antenna Supporting Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

14.8 U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(Current Version) Safety and Health Requirements Manual

14.9 U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction

29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1400	Cranes and Derricks in Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

15.0 DEFINITIONS

15.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

15.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

15.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

15.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

15.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

15.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q, and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented and include experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented, minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

15.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

15.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

15.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

15.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

15.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

15.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

15.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

15.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

15.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the requirements of EM 385-1-1 Appendix Q, and ASSP Z359.0, with a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

15.16 USACE Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

15.17 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap using the Crane High Hazard working group mishap reporting form.

16.0 ACCIDENT PREVENTION PLAN (APP)

16.1. General

A qualified person must prepare the written site-specific APP. The qualified person shall prepare the APP in accordance with the format and requirements of EM 385-1-1, Appendix A, and as supplemented herein. The APP shall cover all paragraph and subparagraph elements in EM 385-1-1, Appendix A. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and must be site-specific. The APP shall describe the methods to evaluate past safety performance of potential subcontractors in the selection process. The APP must also describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. The Contractor is responsible for informing their subcontractors of the safety provisions under the terms of the contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated Site safety and Health Officer (SSHO), the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The Site safety and Health Officer (SSHO) must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

The Contractor shall submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once the APP has been reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. The APP shall be continuously reviewed and amended, as necessary, throughout the life of the contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, Site safety and Health Officer (SSHO) and Quality Control Manager. The Contractor shall incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, the Contractor shall stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. The Contractor shall notify the Contracting Officer within 24 hours of discovery and eliminate and remove the hazard. In the interim, the Contractor shall take all necessary actions to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

16.2 Names and Qualifications

The Contractor shall provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. Qualifications of competent and of qualified persons. At a minimum, the Contractor shall designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

16.2 Plans

The Contractor shall provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

16.2.a **Confined Space Entry Plan**

The Contractor shall develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other Federal, state and local regulatory requirements identified in this contract. The Contractor shall identify the qualified person's name and qualifications, training, and experience. The Contractor shall delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. The Contractor shall include procedures for rescue by contractor personnel and the coordination with emergency responders. (If there isn’t confined space work, include a statement that no confined space work exists and none will be created.)

16.2.b **Standard Lift Plan (SLP)**

The Contractor shall develop a Standard Lif Plan to avoid situations where the operator cannot maintain safe control of the lift. The Contractor shall prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. The Contractor shall maintain the SLP on the LHE for the current lift(s) being made. The Contractor shall maintain historical SLPs for a minimum of 3 months.

16.2.c **Critical Lift Plan - Crane or Load Handling Equipment**

The Contractor shall provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. In addition, Critical Lift Plans are required for the following:

i. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.

ii. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.

iii. For lifts with anticipated binding conditions.

iv. When erecting cranes.

16.2.d. **Critical Lift Plan Planning and Schedule**

Critical lifts require detailed planning and additional or unusual safety precautions. The Contractor shall develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. The Contractor shall comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

16.2.e **Lifts of Personnel**

In addition to the requirements of EM 385-1-1, Section 16.H.02, for lifts of personnel, the Contractor shall demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1, Section 16.T.

16.2.f **Barge Mounted Mobile Crane Lift Plan**

The Contractor shall provide a Naval Architecture Analysis and include an LHE Manufacturer's Floating Service Load Chart in accordance with EM 385-1-1, Section 16.L.03.

16.2.g **Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan**

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. The Contractor shall obtain written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. The Contractor shall demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

16.2.h **Fall Protection and Prevention (FP&P) Plan**

The FP&P Plan must comply with the requirements of EM 385-1-1, Section 21.D and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. The Contractor shall address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 6 feet. A competent person or qualified person for fall protection must prepare and sign the plan documentation. The plan shall include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. The Contractor shall review and revise the Fall Protection and Prevention Plan as necessary and as conditions change, but at a minimum every six months to indicate lengthy projects, changes during the course of construction due to changes in personnel, equipment, systems or work habits. The Contractor shall keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. The Contractor shall include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

16.2.i **Rescue and Evacuation Plan**

The Contractor shall provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Section 21.N and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. The plan shall include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

16.2.j **Hazardous Energy Control Program (HECP)**

The Contractor shall develop a HECP in accordance with EM 385-1-1 Section 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. The Contractor shall submit this HECP as part of the Accident Prevention Plan (APP). The Contractor shall conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. The Contractor shall document the meeting and inspection in accordance with EM 385-1-1, Section 12.A.02. The Contractor shall ensure that each employee is familiar with and complies with the procedures.

16.2.k **Occupant Protection Plan**

The Contractor shall identify the safety and health aspects of lead-based paint removal. This plan shall be prepared in accordance with Section 02 83 00 LEAD REMEDIATION.

16.2.l **Asbestos Hazard Abatement Plan**

The Contractor shall identify the safety and health aspects of asbestos work. This plan shall be prepared in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

16.2.m **Site Safety and Health Plan**

The Contractor shall identify all safety and health aspects related to performing this contract. This plan shall be prepared in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

16.2.n **Polychlorinated Biphenyls (PCB) Plan**

The Contractor shall identify the safety and health aspects of Polychlorinated Biphenyls work associated with this contract. This plan shall be prepared in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

16.2.o **Site Demolition Plan**

The Contractor shall identify all safety and health aspects related to performing demolition for this contract. This plan shall be prepared in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION, along with any accompanying referenced sources.

17.0 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task, or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before being submitted to the Contracting Officer. AHAs must be signed by the Site safety and Health Officer (SSHO), Superintendent, QC Manager and the subcontractor Foreman performing the work. The AHA shall be formatted in accordance with EM 385-1-1, Section 1 or as directed by the Contracting Officer. The Contractor shall submit the AHA for review at least 15 working days prior to the start of each activity, task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

17.1 AHA Management

The Contractor shall review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting. The Contractor shall update the AHA as necessary when procedures, scheduling, or hazards change. The Contractor shall use the AHA during daily inspections by the Site safety and Health Officer (SSHO) to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

17.2 AHA Signature Log

Each employee performing work as part of an activity, task, or DFOW must review the AHA for the specific work that they are performing and they must sign a signature log specifically maintained for that AHA prior to starting work on that activity. The Site safety and Health Officer (SSHO) must maintain a signature log on site for every AHA. The Contractor must provide an interpreter for employees whose primary language is other than English, to ensure a clear understanding of the AHA and its contents.

18.0 DISPLAY OF SAFETY INFORMATION

18.1 Safety Bulletin Board

Within one (1) calendar day after commencement of work, the Contractor shall erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, may be used. The alternative method shall be accessible and include all mandatory information for employee and visitor review. The Contractor shall include and maintain information on the safety bulletin board as required by EM 385-1-1, Section 01.A.07. Additional items required to be posted include:

a. Confined space entry permit.

b. Hot work permit.

18.2 Safety and Occupational Health (SOH) Deficiency Tracking System

The Contractor shall establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. The Contractor shall use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily. The list must provide the following information:

a. Date deficiency identified;

b. Description of deficiency;

c. Name of person responsible for correcting deficiency;

d. Projected resolution date;

e. Date actually resolved.

18.3 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

19.0 EMERGENCY MEDICAL TREATMENT

19.1 General

The Contractor must arrange for their own emergency medical treatment in accordance with EM 385-1-1 current edition. The Government has no responsibility to provide emergency medical treatment.

20.0 NOTIFICATIONS and REPORTS

20.1 Mishap Notification

The Contractor shall notify the Contracting Officer as soon as practical, but no more than twenty-four hours (24hrs), after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Appendix Q, any report of injury, illness, or any property damage. For LHE or rigging mishaps, the Contractor shall notify the Contracting Officer as soon as practical but not more than four hours (4 hrs) after the mishap. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; contract title; type of contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

20.2 Accident Reports

a. The Contractor shall conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. The Contractor shall complete the applicable USACE Accident Report Form 3394, and provide the report to the Contracting Officer within five (5) calendar days of the accident. The Contracting Officer will provide copies of any required or special forms.

b. Near Misses: For Army projects, report all "Near Misses" to the Government Designated Authority (GDA), using local mishap reporting procedures, within 24 hrs. The Contracting Officer will provide the Contractor with the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

c. The Contractor shall conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. The Contractor shall complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within thirty (30) calendar days of the accident. The Contractor shall not proceed with crane operations until the cause of the accident is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form to the Contractor.

20.3 LHE Inspection Reports

The Contractor shall submit the required LHE inspection reports in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

20.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

The Contractor shall provide a FORM 16-1 Certificate of Compliance for LHE entering an activity under this contract and in accordance with EM 385-1-1. The Contractor shall post certifications on the crane.

The Contractor shall develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Section 16.H.03 using Form 16-2 Standard Pre-Lift Crane Plan/Checklist for each lift planned. The Contractor shall submit the SLP to the Contracting Officer for approval within fifteen (15) calendar days in advance of planned lift.

21.0 HOT WORK

21.1 Permit and Personnel Requirements

The Contractor shall submit and obtain a written permit prior from the Fire Division (as required) to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. The Contractor shall provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must have current inspection tags, contain an approved safety pin, and have a tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour (1hr) after completion of the task or as specified on the hot work permit.

When starting work in the facility, the Contractor shall require personnel to familiarize themselves with the location of the nearest fire alarm boxes and to be aware of/know the emergency number of the Fire Division. The Contractor shall REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DIVISION IMMEDIATELY.

21.2 Work Around Flammable Materials

The Contractor shall obtain permit approval from a NFPA Certified Marine Chemist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Section 06.H.

22.0 CONFINED SPACE ENTRY REQUIREMENTS

22.1 General

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

22.2 Entry Procedures

The Contractor shall prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. The Contractor shall comply with EM 385-1-1, Section 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

22.3 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

22.4 Rescue Procedures and Coordination with Local Emergency Responders

The Contractor shall develop and implement an on-site rescue and recovery plan and have accompanying procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

23.0 SEVERE STORM PLAN

23.1 General

In the event of a severe storm warning, the Contractor must:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

24.0 PRODUCTS

24.1 PART 2 PRODUCTS

Not used.

25.0 EXECUTION

25.1 PART 3 EXECUTION

25.2 CONSTRUCTION AND OTHER WORK

The Contractor shall comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

Personal Protective Equipment (PPE) is governed in all areas by the nature of the work the employee is performing. The Contractor and all personnel shall use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn, carried, or available by all personnel. Mandatory PPE includes:

a. Hard Hat

b. Long Pants

c. Appropriate Safety Shoes

c. Appropriate Class Reflective Vests

25.3 Work-Site Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication device(s) must be readily available (easily accessible and within the immediate reach) of the employee. The communication device(s) must be tested prior to the start of work to verify that it effectively operates in the area/environment. An employee check-in/check-out communication procedure must be developed to ensure employee safety.

25.4 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

25.5 Unforeseen Hazardous Material

The Contractor shall identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000) in the contract documents. If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, the Contractor shall stop that portion of work and notify the Contracting Officer immediately. Within fourteen (14) calendar days of receiving notification of hazardous materials, the Government will determine if the material is hazardous. If materials are determined to not be hazardous or not pose danger, the Government will direct the Contractor to proceed without change. If the material is determined to be hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

26.0 UTILITY OUTAGE REQUIREMENTS

The Contractor shall apply for utility outages, in writing, at least 7 calendar days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, the Contractor shall coordinate with the Government on all activities involving the control of hazardous energy.

Applicable activities include but are not limited to:

a. AHAs: The Contractor shall perform a review of HECP and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs).
b. Energized Electrical Circuits: In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits must not be performed without prior government authorization. Government permission is considered through the permit process and submission of a detailed AHA.
c. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

26.1 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shutdown, the Contractor shall conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Section 12.A. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1, Section 12.C. The Contractor shall clarify when personal protective equipment is required during switching operations, inspection, and verification.

26.2 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

The Contractor shall provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Section 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

26.3 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

27.0 FALL PROTECTION PROGRAM

The Contractor shall establish a Fall Protection Program for the protection of all employees exposed to fall hazards. Within the program, the Contractor shall include company policies, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment, and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Sections 21.A and 21.D.

27.1 Training

The Contractor shall institute a Fall Protection Training Program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21.C. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21.C.04 and ASSP Z359.2 in the AHA.

27.2 Fall Protection Equipment and Systems

The Contractor shall enforce use of personal fall protection equipment and systems which include but are not limited to: fall arrest, restraint, and positioning. The fall protection equipment and systems shall be designated for each specific work activity in the Site-Specific Fall Protection and Prevention Plan and AHA, and shall be available at all times when an employee is exposed to a fall hazard. The Contractor shall protect employees from fall hazards as specified in EM 385-1-1, Section 21.

The Contractor shall provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Section 21.I, 29 CFR 1926.500 Subpart M,ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, and ASSP Z359.15.

27.3 Additional Personal Fall Protection

In addition to the required fall protection systems, other protection such as safety skiffs, personal floatation devices, and life rings are required when working above or next to water in accordance with EM 385-1-1, Sections 21.O - 21.O.06. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for ‘tying-off’ in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

27.4 Personal Fall Protection Harnesses

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabiners must be self-closing and self-locking. Snap hooks must be capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. The Contractor must use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. When attaching a person to a fall arrest system, the Contractor must always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall. All full body harnesses must be equipped with Suspension Trauma Preventers such as stirrups, relief steps, or similar additives in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21.I.06.

27.5 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

i. For work within 6 feet of an edge, on a roof having a slope less than or equal to 4:12 (vertical to horizontal), the Contractor shall protect personnel from falling by use of personal fall arrest/restraint systems, guardrails, or safety nets. **A safety monitoring system is not adequate fall protection and is not authorized**. The Contractor shall provide protection to personnel in accordance with 29 CFR 1926.500.

ii. For work greater than 6 feet from an edge, the Contractor shall erect and install warning lines in accordance with 29 CFR 1926.500 and EM 385-1-1, Section L.

b. Steep-Sloped Roofs:

i. Work performed on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, and or safety nets. This requirement also applies to residential or housing type construction.

27.6 Horizontal Lifelines (HLL)

The Contractor shall provide HLL in accordance with EM 385-1-1, Section 21.I.08.d.2. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, and certified prior to use. HLLs shall be used under the supervision of a qualified person for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use, and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site-specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

27.7 Guardrails and Safety Nets

The Contractor shall design, install and use guardrails and safety nets in accordance with EM 385-1-1, Section 21.F.01 and 29 CFR 1926 Subpart M.

27.8 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, the Contractor shall ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. The Contractor shall prepare a Rescue and Evacuation Plan and include a detailed discussion of the following:

a. Methods of rescue
b. Methods of self-rescue or assisted-rescue
c. Equipment used
d. Training requirement
e. Specialized training for the rescuers
f. Procedures for requesting rescue and medical assistance
g. Transportation routes to a medical facility

The Contractor shall include the Rescue and Evacuation Plan as part of the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must comply with the requirements of EM 385-1-1, ASSP Z359.2, and ASSP Z359.4.

28.0 WORK PLATFORMS

28.1 Scaffolding

The Contractor shall provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. The Contractor shall comply with the following requirements:

a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.

c. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.

c. An adequate gate is required.

d. Employees performing scaffold erection and dismantling must be qualified.

d. The scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.

e. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.

g. Special care must be given to ensure scaffold systems are not overloaded.

h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.

i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or another adequate firm foundation.

j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 6 feet.

k. Delineate fall protection requirements when working above 6 feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

28.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the Competent Person (CP). If and when lanyards are used, they must be sufficiently short to prohibit worker from climbing out of the basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying-off to an adjacent pole or structure is not permitted, unless a safe device for 100 percent tie-off is used for the transfer.

Use of Aerial Work Platforms (AWPs) must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of Aerial Work Platforms (AWPs) must be designated as qualified operators by the Prime Contractor. The Contractor shall maintain proof of qualifications on site for review. Proof of qualifications shall also be included in the AHA.

29.0 EQUIPMENT

29.1 Material Handling Equipment (MHE)

a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

b. The use of hooks on equipment purposed for lifting materials must be in accordance with the manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.

c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

29.2 Load Handling Equipment (LHE)

The following requirements apply to LHE:

a. Cranes and derricks must be equipped as specified in EM 385-1-1, Section 16.

b. The Contractor shall notify the Contracting Officer 15 working days in advance of any LHE entering the activity, in accordance with EM 385-1-1, Section 16.A.02, so that necessary quality assurance spot checks can be coordinated. The Contractor's operator must remain with the crane during the spot check. Rigging gear must comply with OSHA, ASME B30.9 Standards and host country safety standards.

c. The Contractor shall comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. The Contractor shall perform erection under the supervision of a designated person (as defined in ASME B30.5). The Contractor shall perform all testing in accordance with the manufacturer's recommended procedures.

d. The Contractor shall comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.

e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Section 11, and ASME B30.5 or ASME B30.22 as applicable.

f. The Contractor shall not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. The Contractor shall not lift personnel with a line hoist or friction crane. Additionally, the Contractor shall submit a specific AHA for this work to the Contracting Officer. The Contractor must ensure the activity and AHA are thoroughly reviewed by all involved personnel.

g. The Contractor shall inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.

h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.

i. The Contractor shall use cribbing when performing lifts on outriggers.

i. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.

j. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.

l. The Contractor shall maintain inspection records in accordance by EM 385-1-1, Section 16.D, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.

m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Section 16.F, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.

n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).

o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.

29.3 Load Handling Equipment (LHE), EXCEPTION

The exception to the above mention LHE requirements are as follows: the aforementioned requirements do not apply to commercial trucks mounted or articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

29.4 Machinery and Mechanized Equipment

a. Proof of qualifications for operator must be kept on the project site for review.

c. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

29.5 Base Mounted Drum Hoists

a. Operation of base mounted drum hoists must comply with EM 385-1-1 and ASSP A10.22.

b. Rigging gear must comply with applicable ASME/OSHA standards.

c. When used on telecommunication towers, base mounted drum hoists must comply with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.

d. AHA must include a written standard operating procedure for Base Mounted Drum Hoists when they are used to hoist personnel. The Operators must have a physical examination in accordance with EM 385-1-1 Section 16.B.05 and must be trained in accordance with EM 385-1-1 Section 16.U and 16.T. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.

e. Material and personnel must not be hoisted simultaneously.

e. The personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.

f. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

PLEASE BE ADVISED: This will be a sample project for offerors. The Government will also add a sample drawing, elevation information, pitch, and additional information to assist Offerors with developing realistic prices for their proposals.

SAMPLE
STATEMENT OF WORK
FOR
ROOF REPAIR AND REPLACEMENT SERVICES
MODIFIED BITUMEN
BUILDING SAMPLE

GENERAL REQUIREMENT

The Contractor shall provide personnel, material, equipment, and supervision to complete the technical requirement in this Statement of Work. The Contractor shall be responsible for hiring labor, equipment vendors and shall follow security and safety directives as explained by the Department of Public Works.

SCOPE OF WORK

1. Completely tear-off and remove all existing flood coat and asphalt/ coal tar build up and insulation from the entire roof. Haul away all job-related debris to a legal disposal site.
2. Clean up roof deck to provide smooth installation surface.
3. Inspect decking and fascia, wood soffit and rake trim for rot or deterioration. Submit Change Order at rates described below for COR approval.
4. Install 50mm thick extruded polystyrene insulation.
5. Apply asphalt primer to entire deck surface at 0.4 liters/square meter and allow to dry
6. Shingle membrane in proper direction to shed water
7. Torch-apply by heating membrane in accordance with manufacturer's recommendation to achieve continuous edge flow and complete bond.
8. Overlap sides minimum 3 inches (75 mm) and end laps minimum 6 inches (150 mm).
9. Extend modified bituminous sheet to 2 inches (50 mm) at perimeter.
10. Extend perimeter multiple ply flashing a minimum of 6 inches (150 mm) onto modified bituminous sheet roofing.
11. Install second ply of modified bitumen in similar fashion. Stagger top ply of modified 18” from bottom ply laps.
12. Where roof accessories are set on modified bituminous sheet roofing, set metal flanges on a secondary sheet of membrane and seal with bead of roofing cement.

13. Layout and position drainage course and allow to lay flat. Cut and fit drainage course to perimeter and penetrations. Overlap adjacent panels at ends and sides so laps of both the core and fabric are in the direction of flow of water.
14. Contractor will fabricate all edge metals.
15. Install new flash pipe collars. (If have vent pipe)
16. Contractor will fabricate all edge metals.
17. Install new flash pipe collars.
18. Keep work area neat at the end of each work day through the duration of the job.
19. Contractor will conduct clean up and remove all equipment and personal from premises.

Direct all questions to the POC or the Project Manager.

<u>Additional Requirements:</u>

a) Contractor shall coordinate the project with the government project manager prior to starting work.

b) Contractor must conform to the Statement of Work and any changes must be approved by the project manager.

c) Contractor must visit the job site with the Government Project Manager and verify site conditions and make measurements prior to bidding the project.

d) All work shall comply with all laws, regulations, and ordinances that pertain to this project.

e) All work associated with this project must be done in compliance with the national building codes.

f) The Contractor will notify the government, in writing, when the project is completed.

g) All necessary inspections are required to be signed off by the Government inspector or the project will not be accepted by the government and payment will not be approved.

h) All items not sized in statement of work or in attachments shall be sized per applicable codes.

i) All work shall comply with the current contract.

j) All material shall be applied or installed in accordance with manufactures recommendations.

k) The Contractor must warranty all parts and labor for a minimum of one year beginning at the time of final Government acceptance.

l) Contractor shall ensure work area is maintained as clean as possible during the project and a complete cleanup at completion of the project.

m) All disturbed walls and surfaces resulting from the demolition or installation of construction related materials shall be restored to match adjacent surfaces.

n) The Contractor must get a dig permit or burn permit, when required, and a copy of the permit must be kept on site while work is being performed.

o) The Contractor must follow all safety guidelines as set forth by OSHA and US Army Corps of Engineers EM185-1-1.

CLIN Structure

1. **Item 01:** Work shall include the removal, disposal (off site), and replacement of metal roof deck. New Metal Roof Deck shall be 18 gauge (Reference UFGS Section 05 30 00). Payment will be based on square feet (SF) of area covered.

2. **Item 02**. Work shall include the removal, disposal (off-site), and replacement of damaged and deficient wood roof substrate sheathing and shall match the existing sheathing dimensions (Reference UFGS 06 10 00). Payment will be based on SF of area covered.

3. **Item 03**. Work shall include the inspection, removal, disposal (off-site), and replacement of damaged or deficient wood roof soffit/eave framing and fascia board members and shall match the existing framing or fascia board dimensions. Payment will be based on board foot.

4. **Item 04.** Work shall include the removal and disposal (off-site) of solid metal soffit covering and/or exposed wood soffit sheathing; the inspection of existing exposed wood soffit framing along the underside of the roof eave; and, the installation of new perforated, continuously vented, finished metal soffit material to match the existing soffit color and finish, along the underside of the roof eave, and/or over existing and/or repaired portions of open wood soffit framing. Reference New metal soffit material color and finish shall match other metal soffit colors and finishes in the same vicinity and from the manufacturer's standard colors and finishes. The Government Contracting Officer Representative (COR) shall review and approve the color and finish through the contract submittal process. Payment will be based on SF of area covered.

5. **Item 05.** Work shall include the removal and disposal (off-site) of building fascia material, inspection of existing wood fascia board members and framing, and the installation of new metal fascia material to match and cover existing wood fascia members (typically 8" to 10" wide – may vary) along the eave or gable wall. New metal fascia material color and finish shall match existing fascia material and shall be from the manufacturer's standard colors and finishes. The Government COR shall review and approve the color and finish through the contract submittal process. Payment will be based on linear feet (LF) of covered area.

6. **Item 06.** Work shall include the removal of shingled roof system materials and/or metal roof system materials and disposal (off-site); complete inspection of the existing substrate; installation of new roofing felts in accordance with UFGS Section 07 61 14.00 20; installation of a new structural standing seam metal roof (SSSMR) system; installation and mechanical fastening of new membrane underlayment over new insulation in accordance with manufacturers specifications to achieve a minimum **105** mile per hour wind resistance; and installation of a new ridge vent system, ventilated eave covers and rake closures. The SSSMR system shall include the ability for the new metal roof to release warm air, which infiltrates into the trapped air space between the new metal roof and the existing roof substrate. New SSSMR material, ridge vents, ventilated eave covers, and rake closure color and finish shall be compatible with and blend and match with color themes and finishes of sloped-roof architectural buildings throughout the Fort Belvoir cantonment area. The new SSSMR material and associated system components' color and finish shall match other METAL ROOF colors and finishes in the same vicinity and from the manufacturer's

standard colors and finishes. The Government COR shall review and approve the color and finish through the contract submittal process. Payment will be based on a single roofing square unit (SQ). Each single roofing SQ equals 100 SF.

7. **Item 07.** Work shall include the installation of new wooden purlins (double layers), bottom layer shall be 2” x 4” Pressure Treated and the top layer shall be 2” x 4” common nailer, perpendicular to the roof slope, along with installation of new wooden nailers over the existing rafters and roof area. Payment will be based on a single roofing SQ. Each single roofing SQ equals 100 SF.

8. **Item 08.** Work shall include the removal, disposal (off-site) of existing Ethylene-Propylene-Diene-Monomer (EPDM) elastomeric membrane roof system, and installation of a Thermoplastic Polyolefin (TPO) membrane roof system. The EPDM roof replacement shall include the following work: removal and disposal (off-site) of all high-density board (if applicable) and roof insulation down to the roof deck substrate (metal roof deck, concrete deck, and/or, wood sheathing deck), including all flashing. Installation of new roof insulation to produce an R-48 value in accordance with UFGS Section 07 22 00; installation and mechanical fastening of new membrane underlayment over new insulation in accordance with manufacturers specifications to achieve a minimum **105** mile per hour wind resistance; installation of new fully adhered 60 mil thick TPO membrane roof system in accordance with UFGS Section 07 13 53 and manufacturer’s recommendations, including new flashing materials, fasteners and closures. *NOTE: Removal of all existing coping material and replacement with new coping material to match the existing coping material shall be ordered under a separate item.* Additional requirements for this item includes installation of a new Retrofit Pro Seal Roof Drain in accordance with manufacturer’s instructions; installation of new tapered insulation to recreate existing and additional crickets, to provide positive drainage of water to all existing roof drains; installation of new wood nailers and wood blocking materials, for perimeter blocking to accommodate the new R-48 insulation; and, installation of new replacement flashing (made of ultra-violet resistant materials) in accordance with the manufacturer’s recommendations and the referenced specifications. The Government COR shall review and approve the Retrofit Pro Seal Roof Drain prior to installation. Existing wood nailers shall **NOT** be reused. The Contractor shall provide new wood

9. **Item 09.** Work shall include the removal, disposal (off-site), and replacement of existing EPDM elastomeric roof system membrane (including membrane underlayment and flashing, as applicable) in accordance with the EPDM Roofing Specification, UFGS Section 07 53 23; installation and mechanical fastening of new membrane underlayment over new insulation in accordance with manufacturers specifications to achieve a minimum **105** mile per hour wind resistance; and, installation of new fully adhered 60 mil thick EPDM roof system in accordance with UFGS Section 07 13 53 and manufacturer’s recommendations, including new flashing materials, fasteners and closures. *NOTE: Removal of all existing coping material and replacement with new coping material to match the existing coping material shall be ordered under a separate line item.* Payment will be based on a single roofing SQ. Each single roofing SQ equals 100 SF.

10. **Item 10.** Work shall include removal, disposal (off-site), and replacement of metal coping on a building roof system or a portion thereof. Replacement shall include new installed coping, minimum 24 gauge, with variable surface dimensions for various buildings throughout the cantonment and/or WSAAF areas, complete with new fasteners and closures as recommended by the manufacturer. New metal coping material color and finish shall be compatible with and blend and match with the existing coping material and with the architectural building themes throughout the Fort Belvoir cantonment area. The new metal

coping color and finish shall match other existing coping material colors and finishes on the building or in the same vicinity and from the manufacturer's standard colors and finishes. The Government COR shall review and approve the color and finish through the contract submittal process. Payment will be based on SF of area covered.

11. **Item 11.** Work shall include the repair of holes or small rips in the TPO roof membrane on an existing roof system. Repair shall be accomplished by placement of a TPO membrane patch. Prior to applying the patch membrane, the area shall be properly cleaned and prepared to receive the patch as per the manufacturer's instructions and recommendations. Patch shall match in thickness and color (white or gray). Payment will be based on SF of area covered.

12. **Item 12.** Work shall include the inspection, removal, disposal (off-site), repair, and replacement of a damaged or deficient portion of an EDPM elastomeric membrane roof system; and removal of insulation to expose the substrate for visual inspection. The repair shall consist of removing the damaged membrane and insulation down to the substrate; inspecting the substrate for damage; installation of new insulation to match the thickness and type of the existing insulation; and repair of the roofing membrane as per the manufacturer's instructions and recommendations in accordance with UFGS Section 07 53 23. Payment will be based on SF of area covered.

13. **Item 13.** Work shall include the repair of holes or small rips in the EPDM roof membrane on an existing roof system. Repair shall be accomplished by placement of an EDPM patch. Prior to applying the patch membrane, the area shall be properly cleaned and prepared to receive the patch as per the manufacturer's instructions and recommendations. Payment will be based on SF of area covered.

14. **Item 14.** Work shall include the repair of base flashing and open seams utilizing cleaner, primer, and nailer and wood perimeter blocking material as required to properly install the finished roofing. Payment will be based on a single roofing SQ. Each single roofing SQ equals 100 SF. cover tape. The repairs shall consist of the proper cleaning of the open seams and base flashing as recommended by the elastomeric membrane manufacturer prior to performing the repairs to the affected areas. Payment will be based on LF of area covered.

15. **Item 15.** Work shall include raising roof mounted equipment, to include but not limited to, Electrical Equipment, Exhaust Hoods, Ventilation Hoods, Smoke Hatches, Lighting Fixtures, and Roof Hatch Doors; and disconnecting and reconnecting electrical wiring by a certified electrician. Maximum unit size is 48" x 48". Plumbing and ductwork are excluded. Payment will be based on per each (EA) unit.

16. **Item 16.** Work shall include the removal and disposal (off-site) of ballasted EPDM elastomeric membrane roof systems down to the roof deck substrate (metal roof deck, concrete roof deck, and/or wood sheathing roof deck) and flashing; installation of new roof insulation that shall produce R-48 value; installation and mechanical fastening of new membrane underlayment over new insulation in accordance with manufacturers specifications to achieve a minimum **105** mile per hour wind resistance; installation of new fully adhered 60 mil thick TPO membrane roof system as specified in UFGS Section 07 13 53 and manufacturer's recommendations, including new flashing materials, fasteners and closures. *NOTE: Removal of all existing coping material and replacement with new coping material to match the existing coping material shall be ordered under a separate line item.* Additional requirements for this item include installation of a new Retrofit Pro Seal Roof Drain in accordance with manufacturer's instructions; installation of new tapered insulation to recreate existing and additional crickets to provide positive

drainage of water to all existing roof drains; installation of new wood nailers and wood blocking materials for perimeter blocking to accommodate the new R-48 insulation; and, installation of new replacement flashing (made of ultra-violet resistant materials) in accordance with the manufacturer's recommendations and the referenced specifications. The Government COR shall review and approve the Retrofit Pro Seal Roof Drain prior to installation. Existing wood nailers shall **NOT** be reused. The Contractor shall provide new wood nailer and wood perimeter blocking material as may be required to properly install the finished roofing. Payment will be based on a single roofing SQ. Each single roofing SQ equals 100 SF.

17. **Item 17.** Work shall include the removal and disposal (off-site) of existing built-up roof system material (consisting of several layers – no ballast) down to the roof deck substrate (metal roof deck, concrete roof deck, and /or wood sheathing roof deck) and flashing; installation of new roof insulation that shall produce R-48 value; installation and mechanical fastening of new membrane underlayment over new insulation in accordance with manufacturers specifications to achieve a minimum **105** mile per hour wind resistance; installation of new fully adhered 60 mil thick TPO membrane roof system as specified in UFGS Section 07 13 53 and manufacturer's recommendations, including new flashing materials, fasteners and closures . *NOTE: Removal of all existing coping material and replacement with new coping material to match the existing coping material shall be ordered under a separate line item.* Additional requirements for this item include installation of a new Retrofit Pro Seal Roof Drain in accordance with manufacturer's instructions; installation of new tapered insulation to recreate existing and additional crickets to provide positive drainage of water to all existing roof drains; installation of new wood nailers and wood blocking materials for perimeter blocking to accommodate the new R-48 insulation; and, installation of new replacement flashing (made of ultra-violet resistant materials) in accordance with the manufacturer's recommendations and the referenced specifications. The Government COR shall review and approve the Retrofit Pro Seal Roof Drain prior to installation. Existing wood nailers shall **NOT** be reused. The Contractor shall provide new wood nailer and wood perimeter blocking material as may be required to properly install the finished roofing. Payment will be based on a single roofing SQ. Each single roofing SQ equals 100 SF.

18. **Item 18.** Work shall include the repair of metal roof mansard knuckles. The repairs for the mansard knuckles will consist of replacing the damaged knuckles with new ones that match the existing knuckles in profile and color. Payment will be based on EA knuckle repaired.

19. **Item 19.** Work is for the repair of damaged metal roof sections. Repairs include replacing/correcting damaged plumbing pipe vents projecting through the roof and installing a watertight metal roof section to match the gauge and color of the existing metal roof. Payment will be based on SF of area covered.

20. **Item 20.** Work shall include disconnection and reconnection of a lightning protection system (LPS) in accordance with UFC 3 575 01. Install air terminals or roof conductors (to include adhesive shoes with adhesive approved by the roof manufacturer) on EPDM, TPO, or SSSMR System roofs in accordance with the roofing manufacturer's instructions and UFGS section 26 41 00. After reconnection of a LPS the Contractor shall provide a UL Lightning Protection Inspection Certificate for the facility, certified by a Lightning Protection Certified Electrician. Payment will be based on LF of the LPS disturbed.

21. **Item 21** Provide 4" rigid insulation, which shall produce an R-24 value. Work shall include the installation of 4" rigid roof insulation composed of a closed cell polyisocyanurate foam core internally laminated (non-asphaltic) and with fiber-reinforced felt facers. Insulation shall be installed as per

manufacturer's instructions and as described in UFGS Section 07 22 00. Payment will be based on SF of area covered.

22. **Item 22.** Provide 4” rigid roof insulation Nail Base (NB). NB is a rigid insulation composite panel composed of a closed cell polyisocyanurate foam core manufactured on-line to a fiber-reinforced facer on one side and 3/4” CDX plywood on the other. This insulation shall produce an R-24 value. It shall be installed as per manufacturer's instructions and as described in Section 07 22 00. The Government COR shall review and approve the NB prior to installation. Payment will be based on SF of area covered.

23. **Item 23.** Work shall include the installation of a Perimeter Enhancement System. Once installation of a TPO membrane roof system or EPDM elastomeric membrane roof system is complete, a Perimeter Enhancement System shall be placed 12” to 24” from the perimeter of the roof. The Perimeter Enhancement System will span the entire perimeter of the roof. The Contractor shall install Batten with 5.5” ultra-ply TPO quick seam flashing per the manufacturer's instructions. The Perimeter Enhancement System offsets increased uplift pressure at the perimeter and corners of the roof and strengthens the roof system. Payment will be based on LF.

24. **Item 24**. Work shall include installation of a new Metal Reglet cut into Masonry or Concrete in accordance with Manufacturers Specifications and UFGS 07 60 00. Payment will be based on LF.

25. **Item 25.** Work shall include the replacement of broken or damaged roof drains. New drains shall come with a dome and integral flange and shall have a device for making a watertight connection between roofing and flashing. The drain assembly shall be galvanized heavy pattern cast iron. Installation drain assemblies shall be in accordance with the drain and roofing manufacturer instructions and UFGS 22 00 00 (2.6.8). Payment will be based on per EA unit.

26. **Item 26**. Work shall include the demolition of existing damaged Snow Guards and the replacement of new Snow Guards. Installation of the Snow Guard assemblies shall be in accordance with the roofing manufacturer instructions. Payment will be based on LF.

27. **Item 27**. Work shall include the Installation of new Snow Guards. Assemblies shall be installed in accordance with the roofing manufacturer instructions. Payment will be based on LF.

28. **Item 28** Work shall include the Installation of Fall Arrest Roof Anchor; Anchor is designed for 5,400 lbs. Ultimate Load. Working load not to exceed 1,800 lbs. Contractor has the responsibility to ensure through its Engineer Consultants that the structure on which this anchor is being attached has sufficient Structural Capacity to safely withstand 5,400 lbs. factored (Ultimate) Load applied on top of an anchor in a direction. Also, the supporting structure must be structurally strong enough to resist a test load of 2,500 lbs. applied at top of anchor, without a permanent deformation of a of its components. Installation of the Fall Arrest Roof Anchor shall be in accordance with the Fall Arrest Roof Anchor manufacturer instructions and their Engineer Consultant. The Government COR shall review and approve the Fall Arrest Roof Anchor prior to installation. Payment will be based on per EA unit.